

Multiple Access Techniques I

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Course Title: Computer Networks



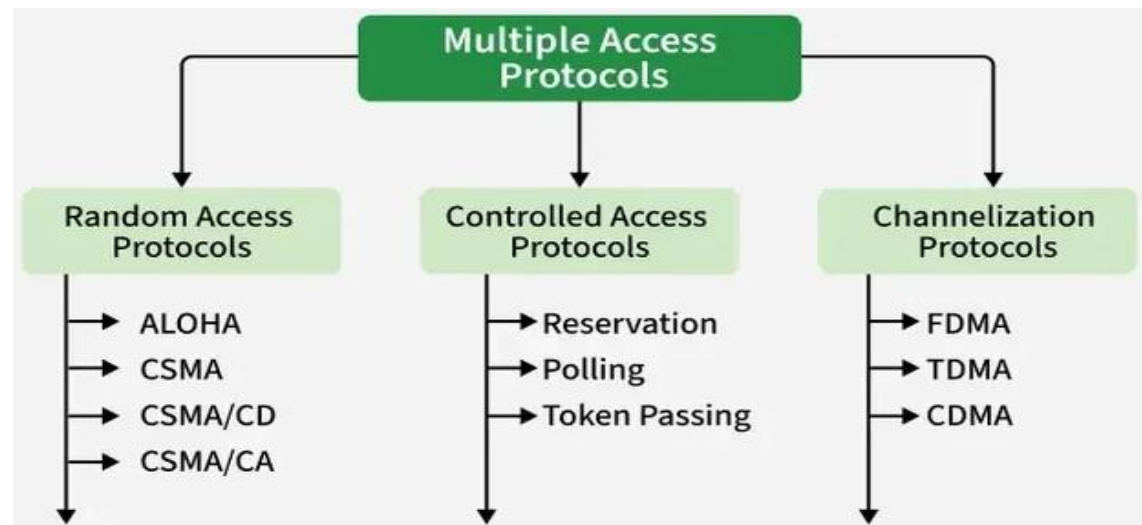
Dept. of Computer Science
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Multiple Access Techniques



- Introduction
- Channelization Protocols
 - FDMA
 - TDMA
 - CDMA
- Controlled Access
 - Reservation
 - Pooling
 - Token-passing

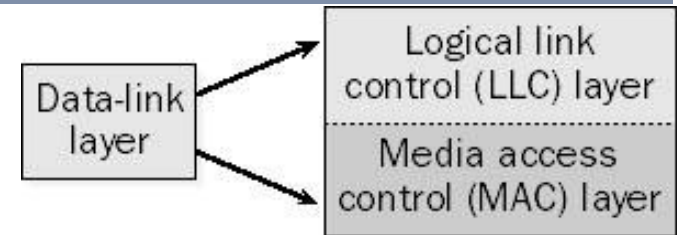


Introduction



► Two sublayers:

1. **LLC (Logical Link Control)** sublayer
2. **MAC (Medium Access Control)** sublayer



- The upper sublayer that is **responsible for flow and error control** is called the logical link control (**LLC**) layer;
- The lower sublayer that is mostly **responsible for multiple access resolution** is called the media access control (MAC) layer. When nodes or stations are connected and use a common link, called a multipoint or broadcast link, we need a multiple-access protocol to coordinate access to the link.

Introduction....

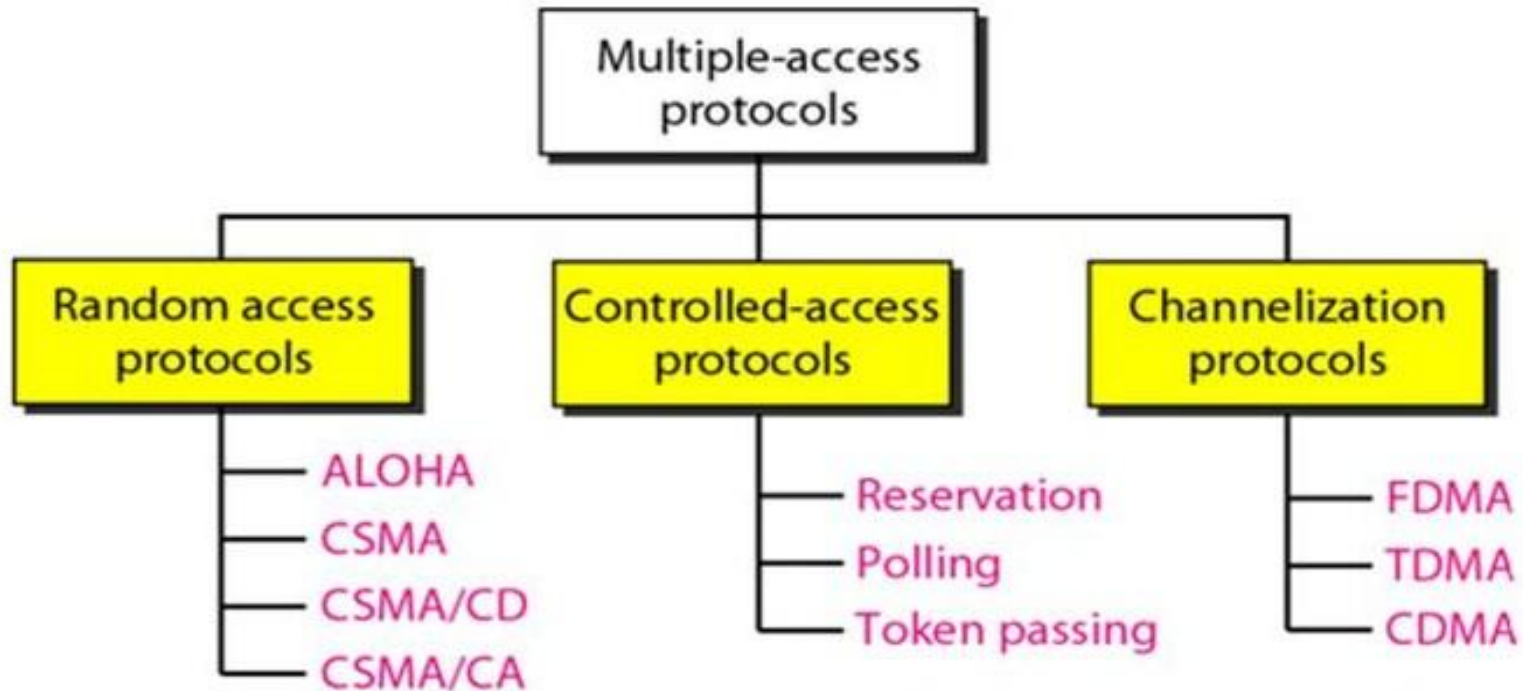


- ▶ Multiple access networks – multiple sending & receiving stations **share the same transmission medium**. Examples: LAN, cellular and satellite networks
- ▶ **Advantages:**
 - ▶ low-cost infrastructure
 - ▶ all stations attached to the medium hear
 - ▶ transmission from any other station $\Rightarrow$ routing not necessary
- ▶ **Disadvantages:**
 - ▶ access of multiple sending and receiving nodes to the shared medium must be coordinated
 - ▶ stations should not be transmitting simultaneously or interrupting each other
 - ▶ stations should not be able to ‘monopolize’ the transmission/shared medium

Introduction....



Classification of multiple access protocols



Channelization Protocols

Classification



- ▶ Channelization is a multiple access method in which the available **bandwidth** of a link is shared in *time*, *frequency* or *code* between different stations.

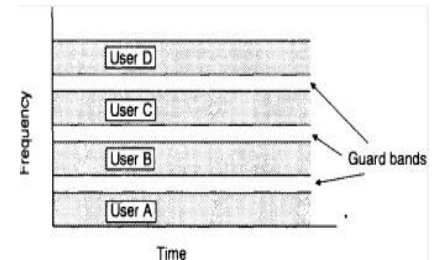
- ▶ There are three basic channelization protocols:
 1. **Frequency Division Multiple Access (FDMA)**
 2. **Time Division Multiple Access (TDMA)**
 3. **Code Division Multiple Access (CDMA)**

Channelization Protocols....

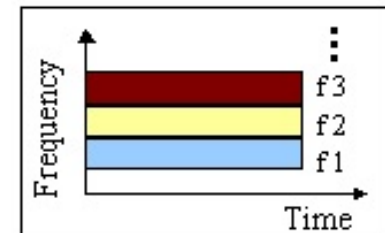
FDMA



- ▶ In FDMA, the available **bandwidth** is divided into **frequency bands**.
- ▶ **Each station** (e.g. f1, f2, user A, user B) is allocated a band to send its data.
- ▶ This **band is reserved** for that station for all the time.
- ▶ The frequency bands of different stations are separated by small bands of unused frequency.
- ▶ These unused bands are called **guard bands** that prevent station interferences.

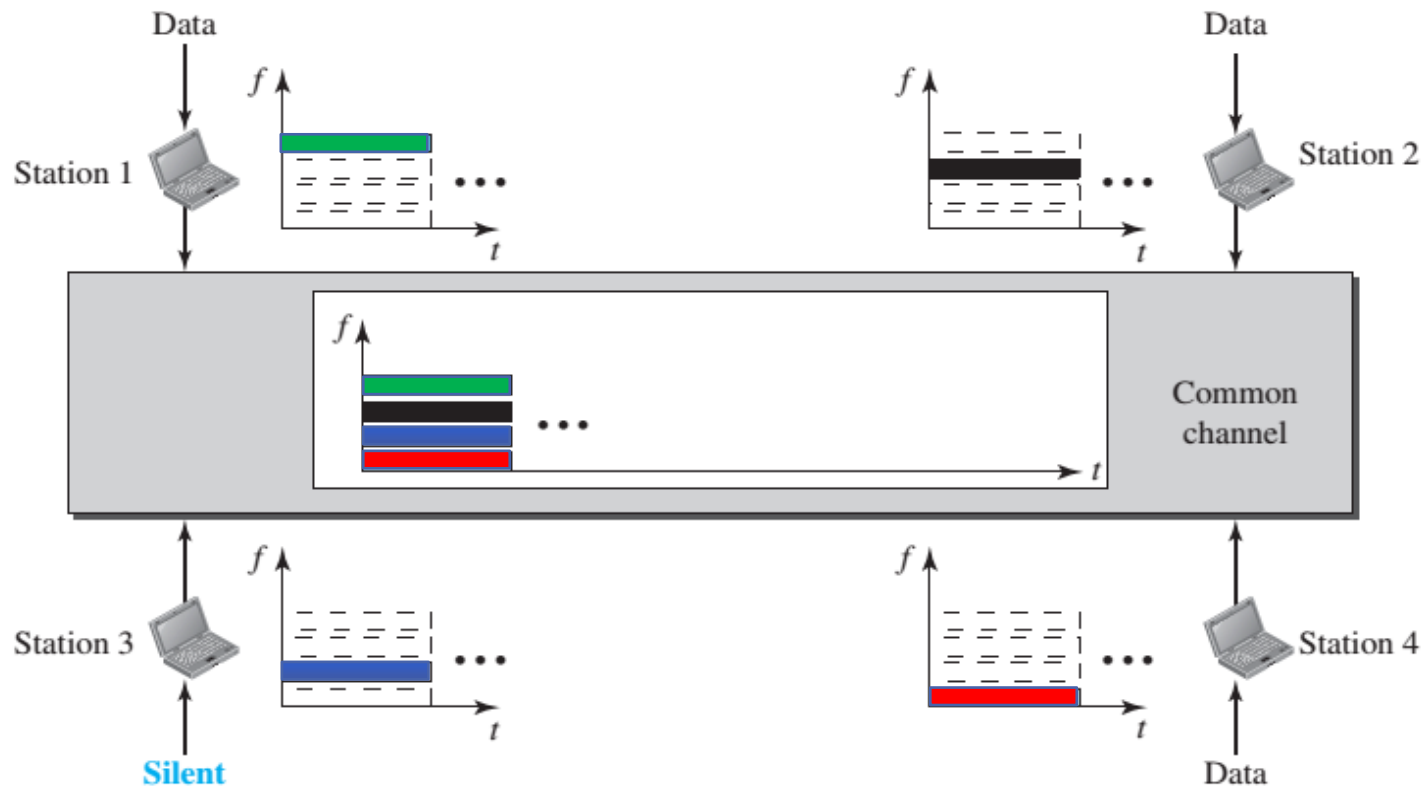


FDMA
(Frequency Division
Multiple Access)



Channelization Protocols....

Illustration of FDMA



Channelization Protocols....

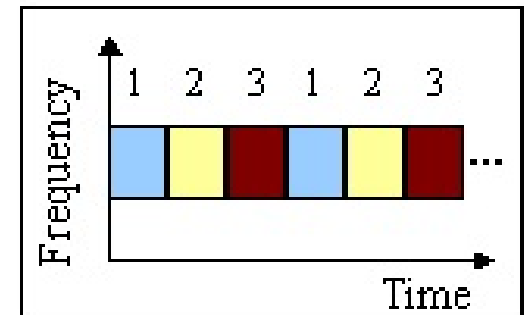
TDMA



- ▶ In TDMA, the bandwidth of channel is **divided** among various stations on the basis of **time**.
- ▶ **Each station** (e.g. 1, 2) is allocated a **time slot** during which it can send its data.
- ▶ Each station must know the beginning of its time slot.
- ▶ TDMA **requires synchronization** between different stations.
- ▶ **Synchronization** is achieved by using some synchronization bits at the beginning of each slot.

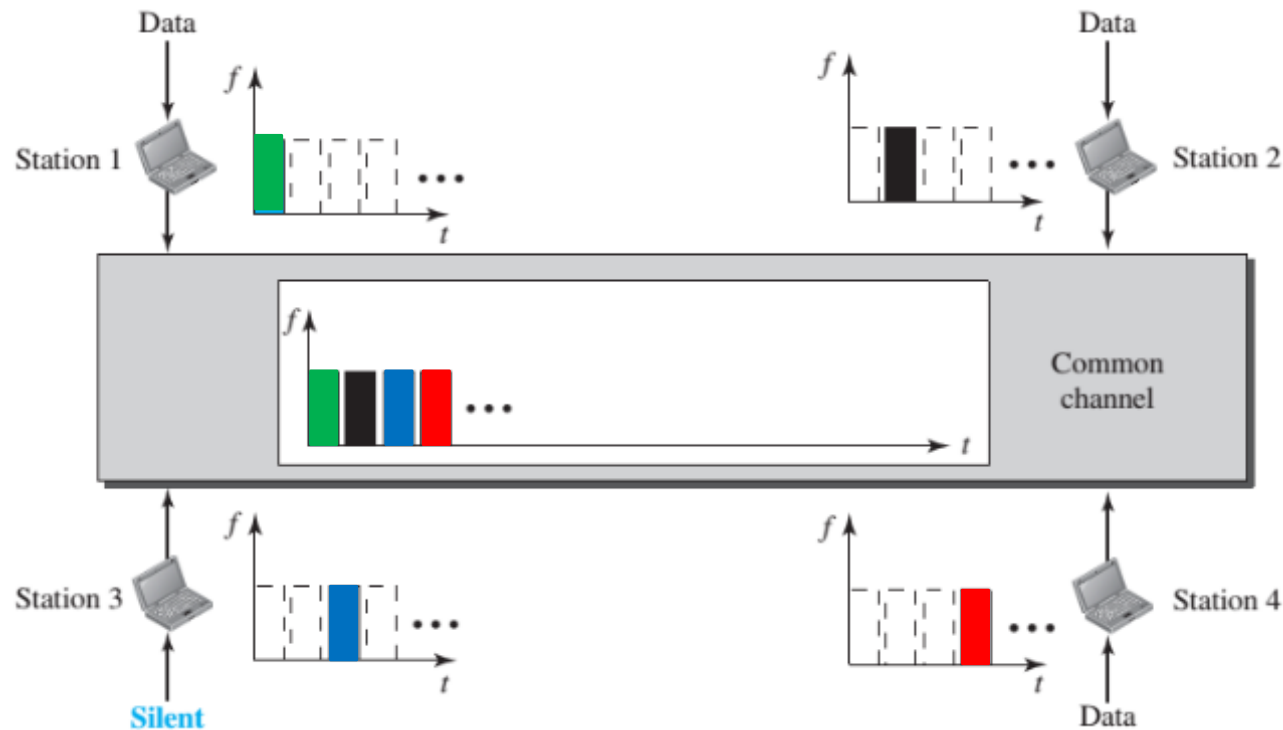
TDMA

(Time Division
Multiple Access)



Channelization Protocols....

Illustration of TDMA



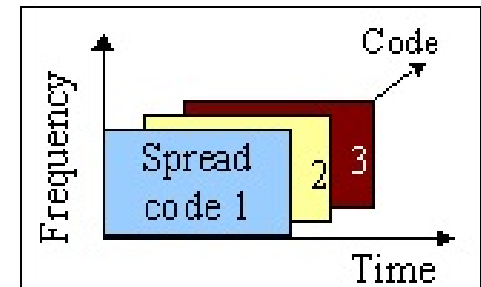
Channelization Protocols....

CDMA



- ▶ Unlike TDMA, in CDMA all stations can **transmit data simultaneously**.
- ▶ CDMA allows each station to transmit over the **entire frequency spectrum** all the time.
- ▶ Multiple simultaneous transmissions are **separated using coding theory**.
- ▶ In CDMA, each user is given a **unique code sequence**.

CDMA
(Code Division
Multiple Access)



Working process of CDMA



- Let us assume that we have 4 stations: 1, 2, 3 and 4; those are connected to the same channel.
- The **data** from **station 1** is **d1**, from **station 2** is **d2** and so on...
- The **code** assigned to **station 1** is **c1**, station c2 and so on.

These assigned codes have two properties:

- ❖ If we multiply each code by another, we get 0, e.g. $c1 \cdot c2 = 0$
- ❖ If we multiply each code by itself, we get 4 (**no of stations**), e.g. $c1 \cdot c1 = 4$

Working process of CDMA

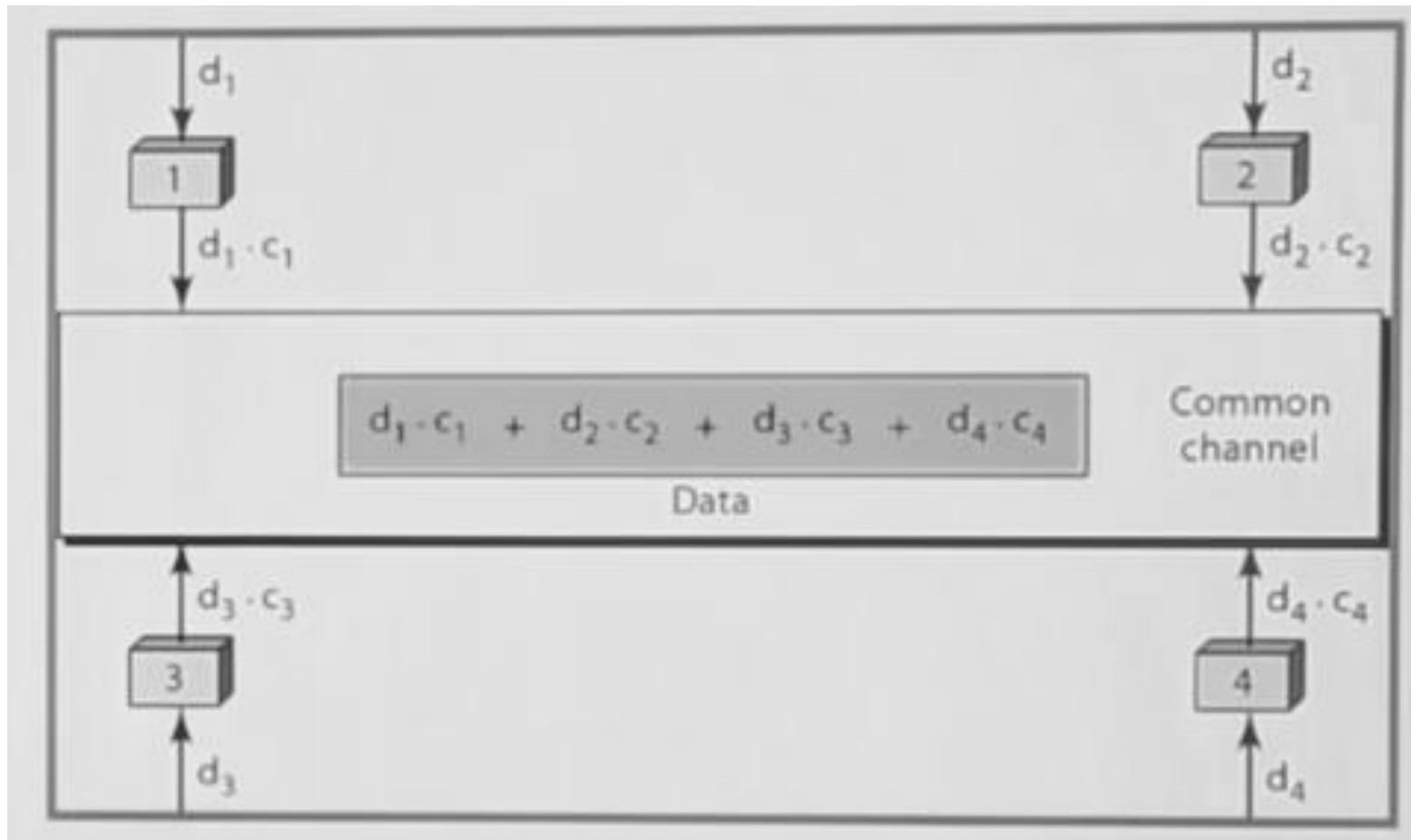
- When these four stations send **data** on the same channel., then station 1 multiplies its data by its **code** i.e. **d1.c1**, station 2 multiplies its data by its code i.e. d2.c2 and so on.....

- The data that goes on the channel is the sum of all these terms:

$$\mathbf{d1.c1} + \mathbf{d2.c2} + \mathbf{d3.c3} + \mathbf{d4.c4}$$

- Any station that wants to receive data from the channel multiplies the data on the channel by the code of the sender.

Working process of CDMA



Working process of CDMA

- For e.g. suppose station 2 wants to receive data from station 1
- It multiplies the **data** on the channel by **c1**, (code of station 1)
- Because (c1, c1) is 4, but (c2, c1), (c3,c1) and (c4,c1) are all 0s, station 2 divides the result by 4 to get the data from station 1.

$$\text{data} = (d1.c1 + d2.c2 + d3.c3 + d4.c4) . \text{c1}$$

$$= d1.c1.c1 + d2.c2.c1 + d3.c3.c1 + d4.c4.c1$$

$$= d1.4 + 0 + 0 + 0$$

$$=(d1.4)/4 = d1$$

Channelization Protocols....

CDMA



❖ How to generate codes?

$$H = [1]$$

$$H^2 = \begin{bmatrix} H^1 & H^1 \\ H^1 & -H^1 \end{bmatrix}$$

$$H^2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H^4 = \begin{bmatrix} H^2 & H^2 \\ H^2 & -H^2 \end{bmatrix}$$

$$H^4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

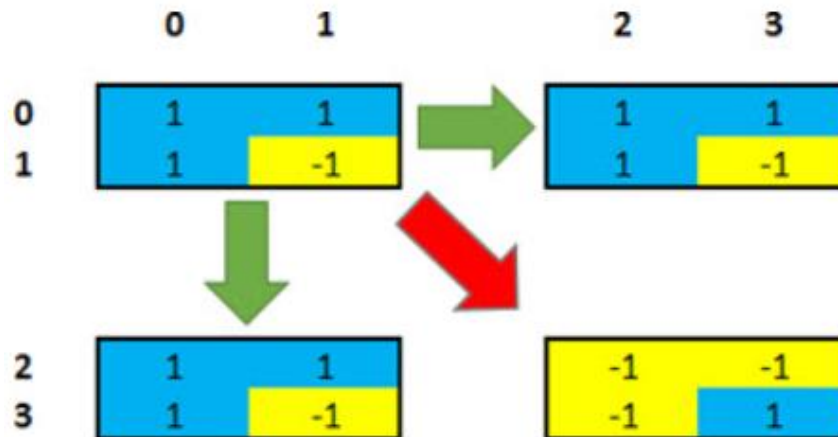
- Each row of H is a code of a user.
- Codes are orthogonal to each other. That is, multiplying any two code results in a zero.

$$H^8 = \begin{bmatrix} H^4 & H^4 \\ H^4 & -H^4 \end{bmatrix}$$

$$H^8 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \end{bmatrix}$$

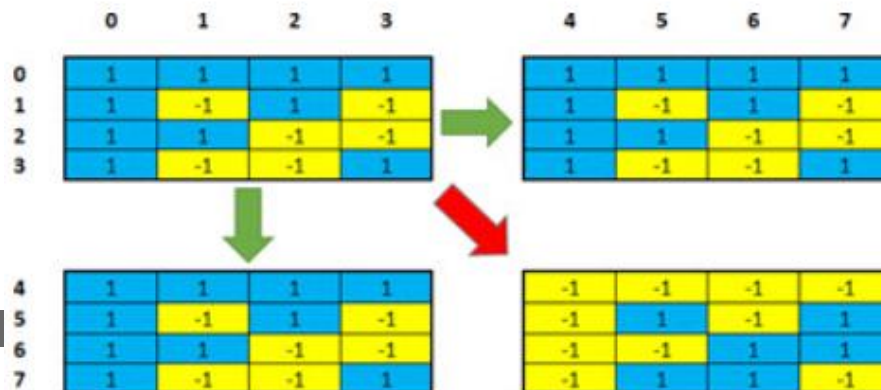
Channelization Protocols....

CDMA



1	1	1	1
1	-1	1	-1
1	1	-1	-1
1	-1	-1	1

Again, we repeat the action, and now we get a 8x8 matrix.



1	1	1	1	1	1	1	1
1	-1	1	-1	1	-1	1	-1
1	1	-1	-1	1	1	-1	-1
1	-1	-1	1	1	-1	-1	1
1	1	1	1	-1	-1	-1	-1
1	-1	1	-1	-1	1	-1	1
1	1	-1	-1	-1	-1	1	1
1	-1	-1	1	-1	1	1	-1

Channelization Protocols....

CDMA



0	1	1	1	1	1	1	1
1	1	-1	1	-1	1	-1	-1
2	1	1	-1	-1	1	1	-1
3	1	-1	-1	1	1	-1	1
4	1	1	1	1	-1	-1	-1
5	1	-1	1	-1	-1	1	1
6	1	1	-1	-1	-1	-1	1
7	1	-1	-1	1	-1	1	-1

i.e. code 2	1	1	-1	-1	1	1	-1	-1
	X	X	X	X	X	X	X	X
i.e. code 5	1	-1	1	-1	-1	1	-1	1
	=	=	=	=	=	=	=	=
	1	-1	-1	1	-1	1	1	-1

$$\Sigma = 0$$

Channelization Protocols....

CDMA



Suppose there are **five** users A, B, C, D and E. They want to send 0, +1, +1, -1, -1 bits, respectively. Generate the code for each user and find the transmit sequence. Suppose that a user G wants to decode the information sent from the user B. How can he do this?

- Since the number of users is 5, use H^8
- Multiply user code by his bit
- Add the results of the multiplication

Channelization Protocols....

CDMA



$$\begin{array}{ll}
 \text{A} & 0 \times R1 = [0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0] \\
 \text{B} & 1 \times R2 = [1 \quad -1 \quad 1 \quad -1 \quad 1 \quad -1 \quad 1 \quad -1] \\
 \text{C} & 1 \times R3 = [1 \quad 1 \quad -1 \quad -1 \quad 1 \quad 1 \quad -1 \quad -1] \\
 \text{D} & -1 \times R4 = [-1 \quad 1 \quad 1 \quad -1 \quad -1 \quad 1 \quad 1 \quad -1] \\
 \text{E} & -1 \times R5 = [-1 \quad -1 \quad -1 \quad -1 \quad 1 \quad 1 \quad 1 \quad 1] \\
 \hline
 & [0 \quad 0 \quad 0 \quad -4 \quad 2 \quad 2 \quad 2 \quad -2]
 \end{array}
 \quad
 \begin{array}{l}
 R1 \text{ (row 1)} \\
 R2 \text{ (row 2)} \\
 \\
 H^8 = \\
 R5 \text{ (row 5)}
 \end{array}
 \begin{bmatrix}
 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\
 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\
 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\
 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\
 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\
 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\
 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1
 \end{bmatrix}$$

Transmit sequence: $[0 \quad 0 \quad 0 \quad -4 \quad 2 \quad 2 \quad 2 \quad -2]$

Channelization Protocols....

CDMA



Recovery of the bit sent from user B

Received sequence $\times$ Desired user's (B's) code

$$\begin{aligned} [0 \quad 0 \quad 0 \quad -4 \quad 2 \quad 2 \quad 2 \quad -2] &\times [1 \quad -1 \quad 1 \quad -1 \quad 1 \quad -1 \quad 1 \quad -1] \\ &= (0 \times 1) + (0 \times -1) + (0 \times 1) + (-4 \times -1) + (2 \times 1) \\ &\quad + (2 \times -1) + (2 \times 1) + (-2 \times -1) \\ &= 0 + 0 + 0 + 4 + 2 - 2 + 2 + 2 \\ &= 8 \end{aligned}$$

The bit sent from user B = $(\text{Received sequence} \times B's \text{ code}) / (\text{length of the code})$
= $8/8 = 1$

Task

CDMA



- ❖ Suppose there are four users A, B, C, and D. They want to send -3, +1, +5, and -1, respectively. Generate the code for each user and find the transmit sequence. Suppose that a user B wants to decode the information sent from the user A. How can he do this?

Controlled Access

Classification



In controlled access, the **stations consult one another to find which station has the right to send.** A station cannot send unless it has been authorized by other stations. We discuss **three popular controlled-access methods.**

1. Reservation
2. Polling
3. Token Passing

Controlled Access....

Reservation



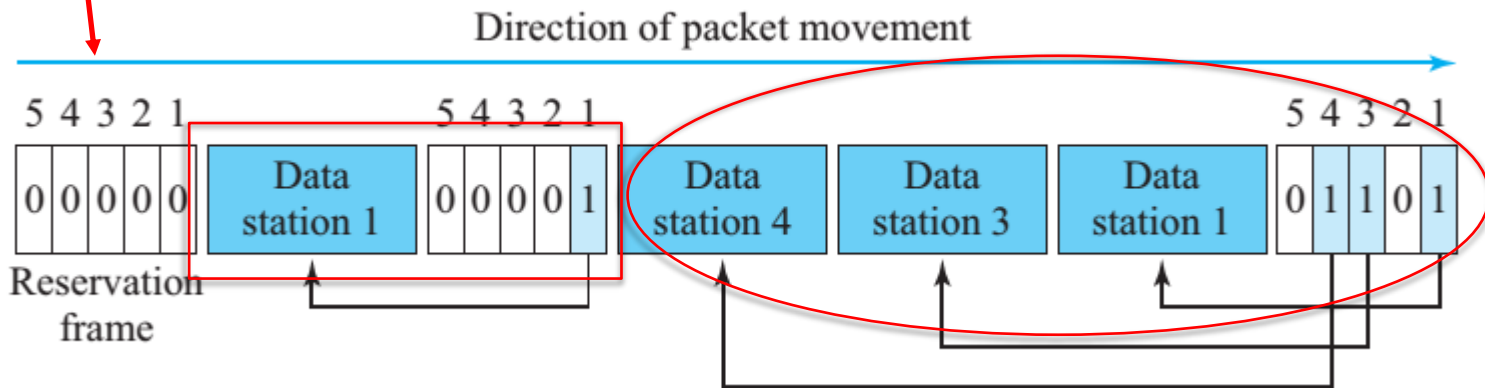
- ▶ In the **reservation method**, a station needs to make a reservation before sending data. Time is divided into intervals. In each interval, a reservation frame precedes the data frames sent in that interval.
- ▶ **How it Works:**
- ▶ If there are N stations in the system, there are exactly N reservation mini lots in the reservation frame. Each mini slot belongs to a station. When a station needs to send a data frame, it makes a reservation in its own mini slot. The stations that have made reservations can send their data frames after the reservation frame.

Controlled Access....

Reservation



In the following figure shows a situation with five stations and a **five-mini slot reservation frame**. In the **first interval**, only stations 1, 3, and 4 have made reservations. In the **second interval**, only station 1 has made a reservation.



Reservation access method

Controlled Access....

Polling



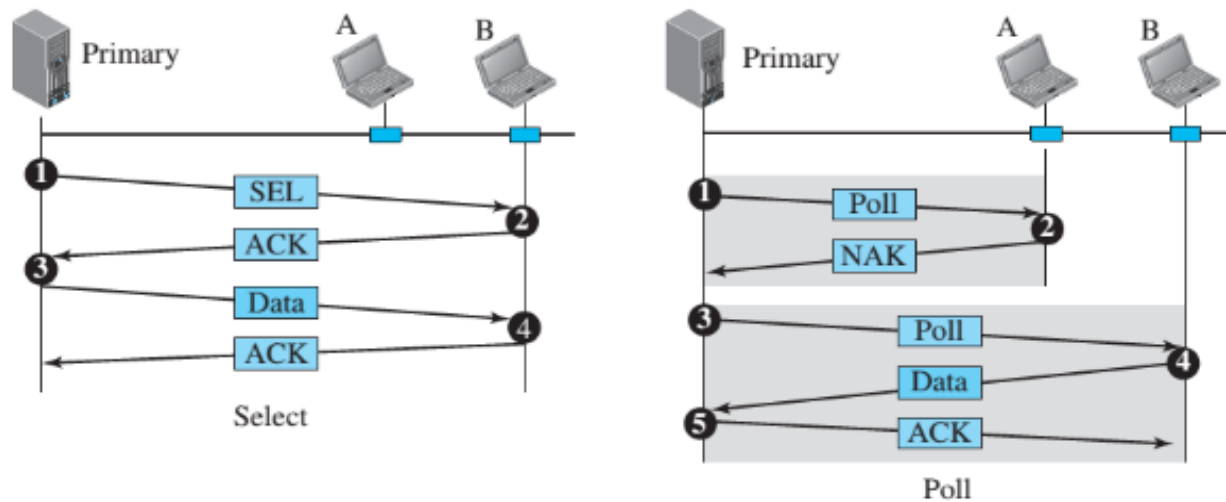
- ▶ Polling works with topologies in which **one device** is designated as a **primary station** and the **other devices** are **secondary stations**.
- ▶ All **data exchanges** must be made **through the primary device** even when the ultimate destination is a secondary device.
- ▶ The **primary device controls the link**; the secondary devices follow its instructions.
- ▶ It is up to the primary device to determine which device is allowed to use the channel at a given time. The **primary device**, therefore, is always the **initiator of a session**.

Controlled Access....

Polling....



If the primary wants to receive data, it asks the secondaries if they have anything to send; this is called **poll function**. If the primary wants to send data, it tells the secondary to get ready to receive; this is called **select function**.



Select and poll functions in polling access method

Controlled Access....

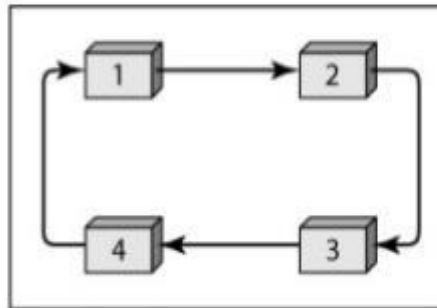
Token-passing method



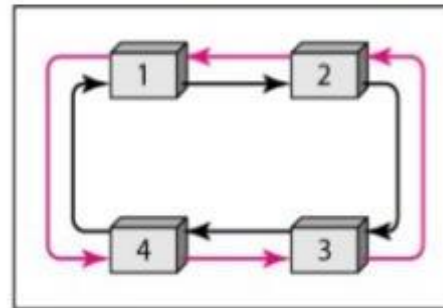
- ▶ In the **token-passing method**, the stations in a **network** are **organized** in a **logical ring**. In other words, for each station, there is a predecessor and a successor. The **predecessor** is the station which is **logically before the station in the ring**; the **successor** is the station which is **after the station in the ring**.
- ▶ **How it works:**
 - ▶ In this method, a **special packet called a token circulates** through the ring. The **possession of the token gives the station the right to access the channel and send its data**.
 - ▶ When a station has some data to send, it **waits until it receives the token from its predecessor**. It then holds the token and sends its data. When the station has no more data to send, it **releases the token**, passing it to the next logical station in the ring.

Controlled Access....

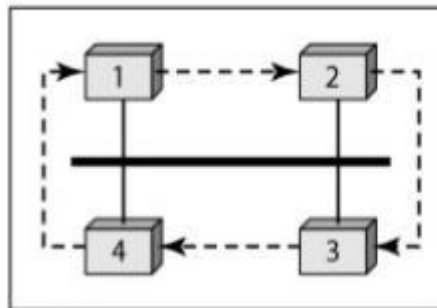
Token-passing method...



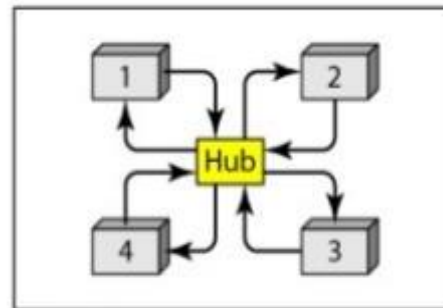
a. Physical ring



b. Dual ring



c. Bus ring



d. Star ring

Logical ring and physical topology in token-passing access method



References

[1] B. A. Forouzan, *Data Communication and Networking*, 5th ed., The McGraw-Hill Companies, Inc., USA, 2013, pp. 341-352.



Recommended Books

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2. **Computer Networking: A Top-Down Approach**, *J. F. Kurose, K. W. Ross*, Pearson Education, Inc., Sixth Edition, USA.
3. **Official Cert Guide CCNA 200-301 , vol. 1**, *W. Odom*, Cisco Press, First Edition, 2019, USA.
4. **CCNA Routing and Switching**, *T. Lammle*, John Wiley & Sons, Second Edition, 2016, USA.
5. **TCP/IP Protocol Suite**, *B. A. Forouzan*, McGraw-Hill, Inc., Fourth Edition, 2009, USA.
6. **Data and Computer Communication**, *W. Stallings*, Pearson Education, Inc., Tenth Edition, 2013, USA.